In PIN photodiodes, an extra high-resistance intrinsic layer is added between p and n layers (Fig. 4). This reduces the transit or diffusion time of photo-induced electron-hole pairs, which, in turn, improves the response time. PIN photodiodes feature low capacitance and therefore high bandwidth, which makes them suitable for high-speed photometry as well as optical communication applications.

Fig. 5: PIN photodiodes in different configurations and packages

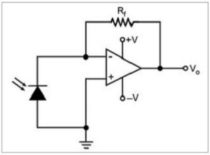

Fig. 6: Application circuit of photodiode in photovoltaic mode

Single-element PIN photodiodes, quadrant photodiodes, and one- and two-dimensional arrays of photodiodes find extensive application in a variety of sensor systems for military applications. Prominent among these include laser warning sensor suites on armoured fighting vehicles and airborne platforms, laser seekers in laser-guided munitions, laser receivers in laser communication systems and focal-plane arrays in Ladar sensors. Photodiodes and photodiode arrays integrated with peripheral components such as low-noise pre-amplifier circuits are also available in a single package.

Fig. 5 shows single, quad and two-dimensional arrays of PIN photodiodes in some of the more common package configurations. Photodiodes are available in many more packages including customised ones. In Schottky type photodiodes, a thin gold coating is sputtered onto the n-material to form a Schottky-effect p-n junction. Schottky photodiodes have enhanced UV response.

Avalanche photodiodes are high-speed, high-sensitivity photodiodes utilising an internal gain mechanism that functions by applying a relatively higher reverse bias voltage than PIN photodiodes. These have fast response times similar to that of PIN photodiodes. Also, responsivity of avalanche photodiodes (40-80A/W) is around 100 times more than silicon PIN photodiodes (0.4-0.6A/W). Moreover, avalanche photodiodes offer an excellent signal-to-noise ratio that is comparable to the value offered by photomultiplier tubes. Hence these are used in a variety of applications requiring high sensitivity, such as long-distance optical communication and optical distance measurement. Military laser rangefinders based on time-of-flight principle have their receivers' front end invariably employing a silicon or indium-gallium-arsenide avalanche photodiode depending upon whether it is an Nd:YAG laser rangefinder or an eye-safe laser rangefinder. Just like PIN photodiodes, avalanche photodiodes are available as single detectors as well as linear or two-dimensional arrays in similar package configurations.

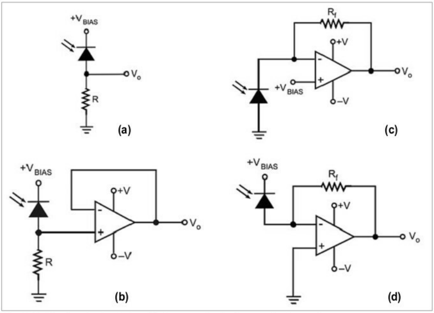

Fig. 7: Application circuits of photodiodes operating in photoconductive mode

Photodiodes can be operated in two modes: photovoltaic and photoconductive. In photovoltaic mode, no bias voltage is applied and due to the incident light, a forward voltage is produced across the photodiode. In photoconductive mode, a reverse bias voltage is applied across the photodiode. This widens the depletion region, resulting in higher speed of response.

As a thumb rule, all applications requiring a bandwidth less than 10kHz can use photodiodes in photovoltaic mode. For all other applications, photodiodes are operated in photoconductive mode. Moreover, the linearity of a photodiode improves when it is operated in photoconductive mode. However, noise current of the photodiode increases when it is operated in photoconductive mode. This is due to the reverse saturation current, referred to as dark current, flowing through the photodiode. The value of dark current is typically in the range of 1-10nA at a specified reverse bias voltage. When the photodiode is operated in photovoltaic mode, dark current is zero.